Kerala Polytechnic REV2021 | Semester 6 | Drawing / Graphics / CADD | Lab

Electrical CAD Lab

Subject Code 6037 lesson HTML for Electrical and Electronics Engineering. This page is a student-friendly study guide with type-specific sections, expected questions and a separate Master Answer Key.

Official syllabus reference: SITTTR course contents | Author: Nandakumar.M | Visit: nandakumarm.dpdns.org

Course Overview

6037 - Electrical CAD Lab

Department(s): Electrical and Electronics Engineering. Semester: Semester 6. Subject type: Lab.

6037

Course Code

Electrical CAD Lab

6

Semester

Semester 6

Draw

Study Mode

Drawing / Graphics / CADD

How to Study

- Read the overview and make a one-page summary.
- Open each module separately and practise the expected questions.
- Keep diagrams, tables, formulas or procedures neat according to subject type.
- Use the Master Answer Key only after attempting the questions.

Study Support

Use Malayalam for classroom discussion if needed, but keep technical terms in English. Attempt module questions first, then check the Master Answer Key.

Drawing Bank

Drawing Bank for Electrical CAD Lab

Quick reference cards for this subject type.

Tools / Software

Drawing sheet, pencils, scales, set squares, compass, mini drafter and CAD commands.

Line Types

Object, hidden, center, dimension, projection and cutting-plane lines.

Sheet Layout

Border, title block, view placement, scale, labels and projection symbol.

Construction Steps

Read problem, draw reference, construct lightly, darken final lines and dimension.

Common Mistakes

Wrong projection placement, missing center lines, poor dimensioning and untidy lettering.

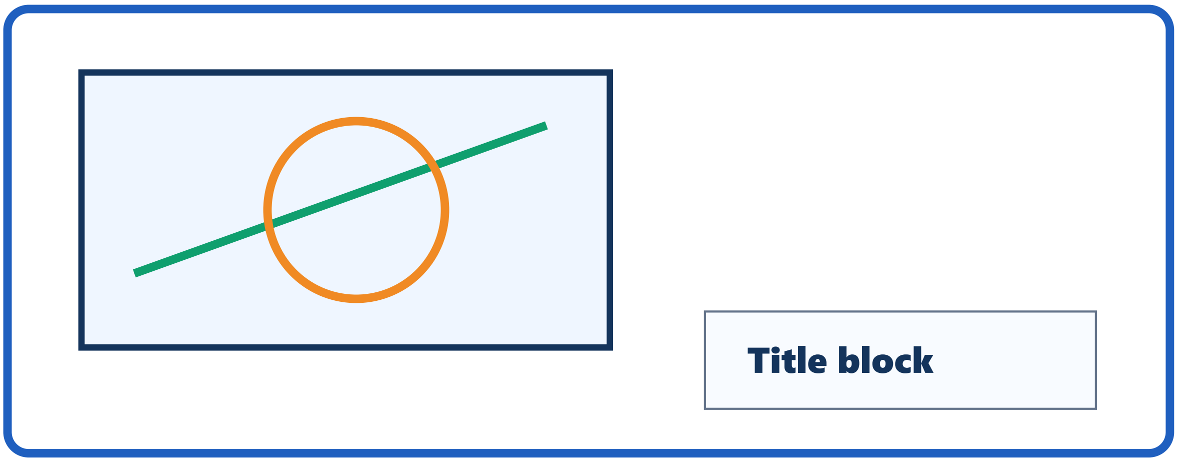
Module 1

Drawing Tools and Standards

Use instruments/software tools, line types and sheet conventions.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Electrical and Electronics Engineering applications.



Visual study block for Electrical CAD Lab

Simple Explanation

Use instruments/software tools, line types and sheet conventions. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

1. List drawing tools/software commands used in this subject.
2. Differentiate object, hidden, center and dimension lines.
3. Write sheet layout rules.
4. List common drawing mistakes.

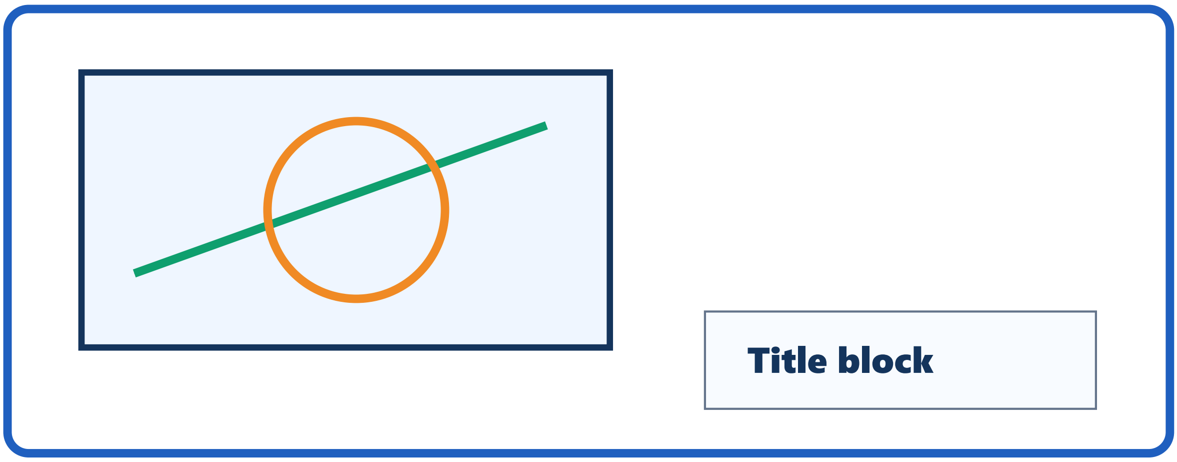
Module 2

Basic Construction and Dimensioning

Practise construction procedure, scale, lettering and dimensions.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Electrical and Electronics Engineering applications.



Visual study block for Electrical CAD Lab

Simple Explanation

Practise construction procedure, scale, lettering and dimensions. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

1. Write the steps for a neat construction problem.
2. Explain dimensioning rules.
3. Prepare a checklist for drawing sheet submission.
4. State the use of scale and projection lines.

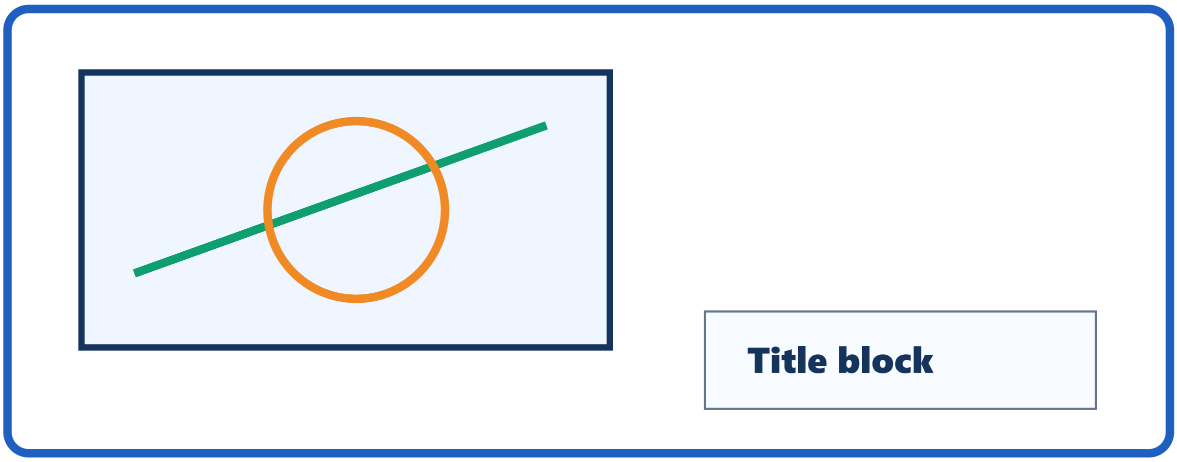
Module 3

Views, Sections and Details

Develop views, sections, layout and labelled drawing presentation.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Electrical and Electronics Engineering applications.



Visual study block for Electrical CAD Lab

Simple Explanation

Develop views, sections, layout and labelled drawing presentation. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

1. Explain view placement for a drawing problem.
2. State the need for sectional view/detail view.
3. Write steps to align multiple views.
4. List labelling rules for final drawing.

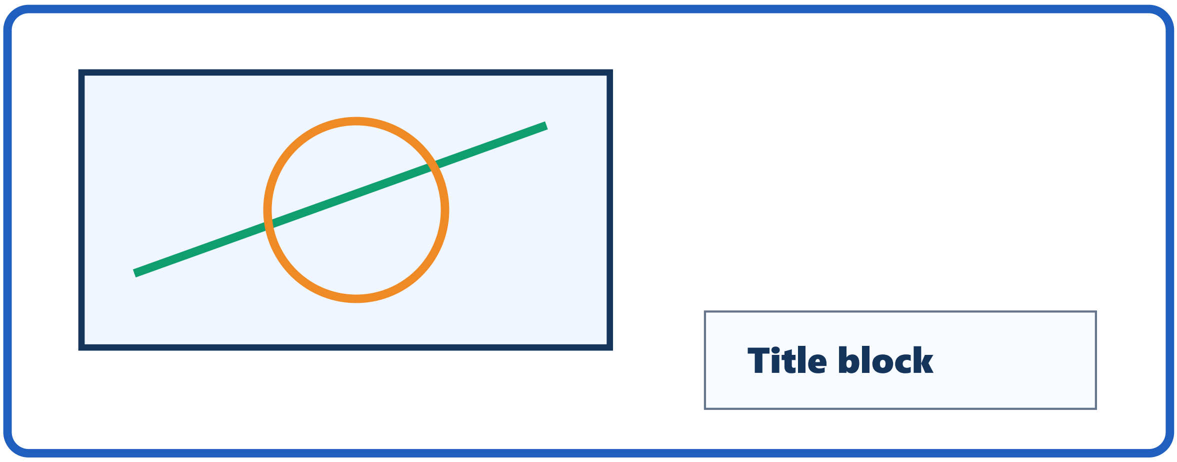
Module 4

CAD / Final Sheet Practice

Use CAD/drawing practice to prepare exam-ready sheets.

Learning Outcomes

- Explain the key ideas in this module.
- Use correct technical terms in answers.
- Prepare neat diagrams/tables/procedures where required.
- Connect the topic with Electrical and Electronics Engineering applications.



Visual study block for Electrical CAD Lab

Simple Explanation

Use CAD/drawing practice to prepare exam-ready sheets. Study the definitions first, then practise diagrams, procedures, tables or numericals according to the subject type.

Study Support

Use Malayalam for classroom discussion if needed, but keep the important technical terms, formulas, symbols, diagrams and final answers in clear English.

Expected Questions

1. List important CAD commands if applicable.
2. Write the final sheet checking method.
3. Explain how to avoid overlapping views.
4. Prepare a viva checklist for drawing/CAD.

Revision

One-page Quick Revision

Last-minute checklist for Electrical CAD Lab.

Important Points

- Definitions and key terms.
- Diagrams, formulas, procedures or formats.
- Applications in Electrical and Electronics Engineering.

Exam Method

OK Read question carefully.

OK Write answer in steps.

OK Label diagrams and units.

OK Finish with result/application.

Common Mistakes

NO Skipping units or labels.

NO Writing answer without structure.

NO Mixing definitions and applications.

NO Leaving diagrams untidy.

Master Answer Key

Answers and Hints

Answer hints exactly match the expected questions inside modules.

Module 1 Answer Hints

M1-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M1-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M1-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M1-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

Module 2 Answer Hints

M2-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M2-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M2-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M2-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

Module 3 Answer Hints

M3-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M3-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M3-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M3-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

Module 4 Answer Hints

M4-Q1

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M4-Q2

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M4-Q3

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

M4-Q4

Answer with construction steps: read given data, draw reference lines, construct lightly, project/align views, darken final lines, dimension and label neatly for Electrical CAD Lab.

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